MIL-F-63987(AR)
26 April 1982
SUPERSEDING
MIL-F-11247A(MU)
(In part)
15 October 1963
(See Section 6)

MILITARY SPECIFICATION

FUZE, MINE, AT, PRACTICE: M604
METAL PARTS FOR

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers quality assurance provisions and preparation for delivery for metal parts for Fuze, Mine, AT, Practice: M604.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification For.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1169	-	packaging, Packing and Marking for Shipment of Inert Ammunition Components.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.5)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9349302	-	Fuze, Mine, Practice, M604 (Metal Parts Assembly).
9349312	-	Cushion.
9349313	-	Cup, Body.
9349314	-	Disc, Body.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 18	-	Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.
ASTM E 117	-	Salt Spray (Fog) Testing.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings, specifications and standards.

3.2 Parts. All parts shall comply with applicable drawings, specifications and standards.

3.3 Mine fuzes. The mine fuzes shall comply with all requirements specified on Drawing (Dwg.) 9349302, all associated drawings, and with all requirements specified in applicable specifications and standards.

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3.4 Load test of cover assembly. The cover assemblies shall withstand a tensile load of not less than 50 pounds without failure when tested as specified in 4.5.4.

3.5 Air pressure test of fuze assembly. The metal parts assemblies shall evidence no drop in pressure when subjected to an internal air pressure of not less than 10 pounds per square inch gage (p.s.i.g.) for not less than three (3) seconds when tested as specified in 4.5.5.

3.6 Pull test of body crimp of fuze assembly. The crimp of the body to the cover assembly shall show no evidence of leakage following a pull of not less than fifteen (15) pounds when tested as specified in 4.5.6.

3.7 Recovery test of firing pin of fuze assembly. The firing pin shall return to a minimum of 0.002 inches above the detonator well shoulder after a load of not less than 134 pounds and not more than 135 pounds is applied and released when tested as specified in 4.5.7.

3.8 Functioning test of fuze assembly. The fuze assembly shall require an axial load of not less than 140 pounds nor more than 240 pounds to cause sudden reversal of the spring when tested as specified in 4.5.8.

3.9 First article inspection. This specification contains technical Provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.10 Workmanship. All parts and assemblies shall be fabricated and assembled in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, chipped edges, gouges, cracks, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents. All required markings shall be neat and sharply defined. External threads shall be full, complete and undamaged for their entire length. All soldering shall be performed in a thorough, workmanlike manner with no evidence of burning or charring.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated below and Table I.

Part Description	Drawings	Quantity
Fuze, Mine, Prac: M604	9349302	25
Body	9349303	10
Firing Pin Assembly	9349304	10
Pin, Firing	9349305	10
Retainer	9349306	10
Spring	9349307	10
Cover Assembly	9349308	10
Cover	9349309	10
Plate, Pressure	9349310	10
Fork, Safety	9349311	10
Cushion	9349312	10
Cup, Body	9349313	10
Disc, Body	9349314	10

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63987(AR)

PARAGRAPH	TITLE		SHEET 1 OF 4		DRAWING NUMBER
	Fuze, Mine, Practice, M604 (Metal Parts Assembly)				See Below
					NEXT HIGHER ASSEMBLY
					--
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
	Fuze, Mine, Practice, M604 (Dwg. 9349302)				
	Examination for Defects	25(a)		3.2	4.4.2.11
	Air pressure test	25(a)		3.5	4.5.5
	Pull test of body crimp	25(a)		3.6	4.5.6
	Recovery test of firing pin	25(a)		3.7	4.5.7
	Functioning test	25(a)		3.8	4.5.8
	Body (Dwg. 9349303)				
	Examination for Defects	10		3.2	4.4.2.1
	Firing Pin Assembly (Dwg. 9349304)				
	Examination for Defects	10		3.2	4.4.2.2

NOTES: (a) Tests shall be performed on the same samples subjected to the examination for defects. Where possible, all tests should be performed on the same twenty-five (25) samples.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63987(AR)

PARAGRAPH	TITLE		SHEET 2 OF 4		DRAWING NUMBER
Cont'd.	Fuze, Mine, Practice, M604 (Metal Parts Assembly)				See Below
					NEXT HIGHER ASSEMBLY --
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Pin, Firing (Dwg. 9349305)				
	Examination for Defects	10		3.2	4.4.2.3
	Salt spray test	(a)		3.2	4.5.1
	Retainer (Dwg. 9349306)				
	Examination for Defects	10		3.2	4.4.2.4
	Salt spray test	(a)		3.2	4.5.1
	Spring (Dwg. 9349307)				
	Examination for Defects	10		3.2	4.4.2.5
	Hardness	(b)		3.2	4.5.2
	Functioning (Snapping) Test	(c)		3.2	4.5.3
	Salt spray test	(a)		3.2	4.5.1

NOTES:

(a) This test shall be performed on the samples following completion of all examination and tests. If any sample shows evidence of corrosion or salt deposits the first article sample shall be rejected.

(b) This test shall be performed on the samples following completion of the examination for defects. If any sample fails to comply with the specified hardness requirements, the first article sample shall be rejected.

(c) This test shall be performed on the samples before performing the salt spray test. If one sample fails to comply with functioning requirements specified on the applicable drawing, the first article sample shall be rejected.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 4		DRAWING NUMBER
Cont'd.	Fuze, Mine, Practice, M604 (Metal Parts Assembly)				See Below
					NEXT HIGHER ASSEMBLY
					--
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
	Cover Assembly (Dwg. 9349308)				
	Examination for Defects	10		3.2	4.4.2.6
	Load test	(a)		3.4	4.5.4
	Cover (Dwg. 9349309)				
	Examination for Defects	10		3.2	4.4.2.7
	Plate, Pressure (Dwg. 9349310)				
	Examination for Defects	10		3.2	4.4.2.8
	Salt Spray test	(b)		3.2	4.5.1

(a) This test shall be performed on the samples subjected to the examination for defects. If any sample fails to comply with the specified requirements, the first article sample shall be rejected.

(b) This test shall be performed on samples subjected to the examination for defects. If any sample shows evidence of corrosion or salt deposits, the first article sample shall be rejected.

NOTES:

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63987(AR)

PARAGRAPH	TITLE		SHEET 4 OF 4		DRAWING NUMBER
Cont'd.	Fuze, Mine, Practice, M604 (Metal Parts Assembly)				See Below
					NEXT HIGHER ASSEMBLY --
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Fork, Safety (Dwg. 9349311)				
	Examination for Defects	10		3.2	4.4.2.9
	Salt spray test	(a)		3.2	4.5.1
	Cushion (Dwg. 9349312)				
	Inspect for all drawing requirements	10		3.2	N/A
	Cup, Body (Dwg. 9349313)				
	Examination for Defects	10		3.2	4.4.2.10
	Salt spray test	(a)		3.2	4.5.1
	Disc, Body (Dwg. 9349314)				
	Inspect for all drawing requirements	10		3.2	N/A

NOTES: (a) This test shall be performed on samples subjected to the examination for defects. If any sample shows evidence of corrosion or salt deposits, the first article sample shall be rejected.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the classification of Defects and Tests Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63987(AR)

PARAGRAPH	TITLE	SHEET 1 OF 3	DRAWING NUMBER
4.4.2.1	Body		9349303
			NEXT HIGHER ASSEMBLY 9349302

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of primer cavity		0.40%(a)	3.2	Gage
102	Depth of primer cavity (.365)		0.40%(a)	3.2	Gage
103	Height of primer cavity crimp projection (.025)		0.40%(a)	3.2	Gage
104	Radius at shoulder bottom of primer cavity		0.40%(a)	3.2	Gage
105	Diameter of firing pin hole above primer cavity		0.40%(a)	3.2	Gage
106	Diameter of spring counterbore (.977)		0.40%(a)	3.2	Gage
107	Radius to start of spring taper, max		0.40%(a)	3.2	Gage
108	Depth to start of spring taper (.035)		0.40%(a)	3.2	Gage
109	Diameter of counterbore above firing pin hole (.85)		0.40%(a)	3.2	Gage
110	Width of primer cavity crimp projection, max (.010)		0.40%(a)	3.2	Gage
111	Distance between bottom cover counterbore and bottom of counterbore above firing pin hole (.17)		0.40%(a)	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63987(AR)

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
4.4.2.1 Cont'd.	Body				9349303
					NEXT HIGHER ASSEMBLY 9349302
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
112	Distance between bottom cover counterbore and bottom of primer cavity		0.40%(a)	3.2	Gage
113	Distance between body cup seat shoulder and bottom of body (.50)		0.40%(a)	3.2	Gage
114	Diameter of cover counterbore		0.40%(a)	3.2	Gage
115	Depth of cover groove		0.40%(a)	3.2	Gage
116	Outside diameter at top of body		0.40%(a)	3.2	Gage
117	Outside diameter under body cup seat shoulder		0.40%(a)	3.2	Gage
118	Outside diameter bottom of body (.976)		0.40%(a)	3.2	Gage
119	Width of cover groove (.010)		0.40%(a)	3.2	Gage
120	Total height		0.40%(a)	3.2	Gage
121	Concentricity of spring counterbore with firing pin hole		0.40%(a)	3.2	Gage
122	Chamfer to firing pin hole missing		0.40%(a)	3.2	Visual
123	Casting workmanship poor (holes, cracks, shrinkage, incomplete fillets or recesses)		0.40%(a)	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63987 (AR)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
4.4.2.1 Cont'd.	Body				9349303
					NEXT HIGHER ASSEMBLY 9349302
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Cavity, hole, or counterbore finish improper		0.40%(a)	3.2	Visual
202	Poor workmanship (excl. 123 above)		1.0%	3.10	Visual

NOTES: (a) See 4.4.3.1 for alternate method of sampling.

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Quality conformance inspection

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Firing Pin Assembly				9349304
					NEXT HIGHER ASSEMBLY
					9349302
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Height to cocked springs		0.40%	3.2	Gage
102	Crimp not 360 degrees		0.40%	3.2	Visual
103	Springs incorrectly assembled		0.40%	3.2	Visual
104	Firing pin point damaged (chipped, mushroomed)		0.40%	3.2	Visual
105	Spring not in assembly position (prior to insertion into fuze)		0.40%	3.2	Visual
Minor					
201	Lacquer missing or inadequate		0.65%	3.2	Visual
202	Poor workmanship		1.0%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63987(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Firing Pin				9349305
					NEXT HIGHER ASSEMBLY 9349304
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Profile of point (includes length, angle, diameter and radius)		0.40%(a)	3.2	Gage
102	Diameter of shank above point		0.40%(a)	3.2	Gage
103	Length to largest outside diameter from point		0.40%(a)	3.2	Gage
104	Diameter of retainer mating surface (.216)		0.40%(a)	3.2	Gage
105	Length of retainer mating surface (.10)		0.40%(a)	3.2	Gage
106	Diameter of head (largest outside diameter)		0.40%(a)	3.2	Gage
107	Width of head		0.40%(a)	3.2	Gage
108	Total length		0.40%(a)	3.2	Gage
Minor					
201	Diameter of boss on head (.26)		0.65%(a)	3.2	Gage
202	Finish improper		0.65%	3.2	Visual
203	Poor workmanship		1.0%	3.10	Visual

NOTES: (a) See 4.4.3.2 for alternate method of sampling.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Retainer				9349306
					NEXT HIGHER ASSEMBLY
					9349304
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Total height		0.40%(a)	3.2	Gage
102	Length to taper		0.40%(a)	3.2	Gage
103	Angle of taper under flange		0.40%(a)	3.2	Gage
104	Diameter of flange		0.40%(a)	3.2	Gage
105	Smaller outside diameter		0.40%(a)	3.2	Gage
106	Inside diameter		0.40%(a)	3.2	Gage
107	Countersink missing or inadequate		0.40%(a)	3.2	Visual
Minor					
201	Finish improper		0.65%(a)	3.2	Visual
202	Poor workmanship		1.0%	3.10	Visual

NOTES: (a) See 4.4.3.2 for alternate method of sampling.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	pring (after heat treatment)				9349307
					NEXT HIGHER ASSEMBLY
					9349304
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Hardness		(a)	3.2	4.5.2
102	Functioning (Snapping) test		(b)	3.2	4.5.3
103	Outside diameter		0.40%(c)	3.2	Gage
104	Hole diameter		0.40%(c)	3.2	Gage
105	Concentricity of outside diameter with hole		0.40%(c)	3.2	Gage
106	Thickness		0.40%(c)	3.2	Gage
107	Height		0.40%(c)	3.2	Gage
108	Identification mark missing, improperly located or color incorrect		0.40%(c)	3.2	Visual
Minor					
201	Poor workmanship		1.0%	3.10	Visual

(a) Five (5) samples shall be selected from each heat treatment batch and subjected to the hardness test. If any sample fails to comply with the hardness requirement specified, the heat treat batch shall be rejected.

(b) Sampling shall be in accordance with MIL-STD-105 using an AQL of 1.00 percent and Code letter H. If any spring fails to comply wih the specified requirements, it shall be classed defective.

(c) See 4.4.3.3 for alternate method of sampling.

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Cover Assembly				9349308
					NEXT HIGHER ASSEMBLY 9349302
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Load test	(a)		3.4	4.5.4
102	Distance between plate and cover		0.40%	3.2	Gage
103	Solder extends beyond specified portion of cover		0.40%	3.2	Visual
104	Solder workmanship unsatisfactory		0.40%	3.2	Visual
105	Assembly incorrect		0.40%	3.2	Visual
106	Plate or cover damaged (crack, puncture, deformation)		0.40%	3.2	Visual
Minor					
201	Protective coating on plate damaged (bare spot exposing base metal)		0.65%	3.2	Visual
202	Marking on plate incorrect, misleading, or missing		0.65%	3.2	Visual
203	Poor workmanship		1.0%	3.10	Visual

NOTES: (a) Sampling shall be in accordance with MIL-STD-105 using a AQL of 1.50% and Code letter G. If any cover assembly fails to comply with the specified requirements, it shall be classed defective.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Cover				9349309
					NEXT HIGHER ASSEMBLY
					9349308
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Height to plate seat surface		0.40%(a)	3.2	Gage
102	Height of plate dimple		0.40%(a)	3.2	Gage
103	Outside diameter		0.40%(a)	3.2	Gage
104	Width of flange (.04)		0.40%(a)	3.2	Gage
105	Thickness of flange		0.40%(a)	3.2	Gage
106	Radius between plate dimple and plate seat surface		0.40%(a)	3.2	Gage
107	Radius missing		0.40%(a)	3.2	Gage
Minor					
201	Poor workmanship		1.0%	3.10	Visual

NOTES:

(a) See 4.4.3.3 for alternate method of sampling.

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Plate, Pressure				9349310
					NEXT HIGHER ASSEMBLY
					9349308
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Flange thickness		0.40%(a)	3.2	Gage
102	Height of dimple		0.40%(a)	3.2	Gage
103	Outside radius start of dimple (.07)		0.40%(a)	3.2	Gage
Minor					
201	Outside diameter		0.65%(a)	3.2	Gage
202	Marking incorrect, misleading or missing		0.65%	3.2	Visual
203	Protective coating damaged (bare spot exposing base metal)		0.65%	3.2	Visual
204	Poor workmanship		1.0%	3.10	Visual

NOTES: (a) See 4.4.3.3 for alternate method of sampling.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63987(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Fork, Safety				9349311
					NEXT HIGHER ASSEMBLY 9349302
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness		0.40%(a)	3.2	Gage
102	Inside diamter closed end		0.40%(a)	3.2	Gage
103	Larger inside diameter		0.40%(a)	3.2	Gage
104	Width of opening		0.40%(a)	3.2	Gage
Minor					
201	Distance between indicated diameters		0.65%(a)	3.2	Gage
202	Radii, max (4 places)		0.65%(a)	3.2	Gage
203	Width (.187)		0.65%(a)	3.2	Gage
204	Protective coating damaged (bare spot exposing base metal)		0.65%	3.2	Visual
205	Poor workmanship		1.0%	3.10	Visual

NOTES: (a) See 4.4.3.3 for alternate method of sampling.

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10	Cup, Body				9349313
					NEXT HIGHER ASSEMBLY
					--
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Total length		0.40%(a)	3.2	Gage
102	Inside diameter		0.40%(a)	3.2	Gage
103	Thickness at bottom		0.40%(a)	3.2	Gage
104	Diameter of hole		0.40%(a)	3.2	Gage
105	Radius, max (bottom)		0.40%(a)	3.2	Gage
106	Outside diameter, max		0.40%(a)	3.2	Gage
Minor					
201	Protective coating damaged (bare spot exposing base metal)		0.65%	3.2	Visual
202	Poor workmanship		1.0%	3.10	Visual

NOTES: (a) See 4.4.3.3 for alternate method of sampling.

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.11	Fuze, Mine, Practice: M604				9349302
					NEXT HIGHER ASSEMBLY
					--
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Air pressure test		100%	3.5	4.5.5
102	Pull test of body crimp	(a)		3.6	4.5.6
103	Recovery test of firing pin	(a,b)		3.7	4.5.7
104	Functioning test	(a,c)		3.8	4.5.8
105	Total height, max		0.40%	3.2	Gage
106	Distance between pressure plate flange and top of body, min		0.40%	3.2	Gage
107	Distance between bottom of detonator cavity and tip of firing pin, min		0.40%	3.2	Gage

a. A sample of thirty-two (32) fuzes shall be selected from each lot for this test. If two (2) or more fuzes fail to comply with the specified requirements, the lot shall be rejected. If no defect is found the lot shall be accepted. If one (1) defect is found a second sample of thirty-two (32) fuzes may be tested at the Contractor's request. If the combined number of defects in both the first and second samples is two (2) or more, the lot shall be rejected.

(b) Fuzes subjected to the recovery test may be used in the functioning test or scrapped.

(c) Fuzes subjected to the functioning test shall be scrapped.

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9349302
4.4.2.11 Cont'd.	Fuze, Mine, Practice: M604				NEXT HIGHER ASSEMBLY --
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
108	Part improperly assembled		0.40%	3.2	Visual
109	Part damaged		0.40%	3.2	Visual
110	Part missing		0.40%	3.2	Visual
Minor					
201	Poor workmanship		1.0%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-63987(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Carton (Sealed)				See Section 5
					NEXT HIGHER ASSEMBLY --
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Contents move when carton is shaken		0.40%	3.2	Manual
102	Carton damaged so that contents are exposed or liable to become exposed		0.40%	3.2	Visual
103	Sealing strip incomplete, badly wrinkled or not in contact with surface for at least 1/2 inch around entire carton		0.40%	3.2	Visual
Minor					
201	Marking misleading, unidentifiable, or missing		0.65%	3.2	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First article and Quality conformance inspection tables.

4.4.3.1 Dimensional control of cast parts. In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least three (3) parts (as cast) from each cavity may be dimensionally inspected to qualify a new or reworked cavity for use in production or as a control of the cavity during production. Inspection for control of each cavity during production shall be performed on the above quantities from each cavity after production of 5000 parts. Individual cavity identification must be provided.

If any defective parts are found during qualification of the cavity, the cavity producing the defective part shall not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for corrective action.

All cavities formerly removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test as above. The contractor may request a change of inspection frequency providing he presents objective evidence to substantiate the request to the Contracting Officer.

4.4.3.2 Alternate method of inspection of dimensional characteristics of retainer and firing pin. In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) of each part shall be dimensionally inspected to qualify the machine used in the fabricating of the part for use in production. In addition, a random sample of five (5) of each part shall be selected from the last portion of each hour's production for dimensional inspection as a control of the machine during production.

If any defective parts are found during qualification of the machine, the machine producing the defective part shall not be used in production until adjustments or rework of the machine are made to produce acceptable parts.

If any defective parts are found when inspection is performed for control of the machine, the machine producing the defective part shall be removed from production, Further, that portion of production since the last machine control check shall be returned to the contractor for corrective action.

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All machines removed from production because of some fault, may, after reworking, or readjustment, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.4.3.3 Dimensional control of stamped parts. In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts may be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.4.4 Inspection Equipment. The inspection equipment required to perform the inspections and tests Prescribed in this specification is identified in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1, and the test method paragraph (See 4.5). The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Salt spray test. The salt spray test shall be performed utilizing the equipment, methods and procedures specified in ASTM B-117 and in accordance with the applicable detailed specification for the protective finish. Exposure time shall be not less than twenty-four (24) hours using an atomized sodium chloride solution of five percent (5%). Unless otherwise specified, this test shall be performed on first article samples only.

4.5.2 Hardness. This test shall be performed as specified in ASTM E 18. Each heat treat batch shall remain segregated until completion of all required tests.

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4.5.3 Functioning (snapping) test of spring. The spring shall be subjected to an axial load at the rate of 1/2 to 1 inch per minute to cause it to snap through and then to snap back. Observation shall be made that the reversed height is 0.068 plus .010 inch, a load of 60 to 90 pounds is required to snap back, the travel during snap back is 0.032 plus .015 inch, the height after snap back is 0.070 inch, minimum and no enamel is flaked off during the testing to determine compliance with the requirements specified on the applicable drawing. Springs so tested shall be scrapped.

4.5.4 Load test of cover assembly. The cover assembly shall be subjected to a tensile load applied between the pressure plate and the fully supported cover. The load shall be applied at such a rate that the pull fixture moves 1/2 to 1 inch per minute until the solder joint is destroyed. Observation shall be made for a failure load of fifty (50) pounds, minimum, and evidence of 360 degree solder contact in the soldered joint for compliance with 3.4.

4.5.5 Air pressure test of fuze assembly. Each fuze assembly shall be subjected to an internal air pressure of ten (10) pounds per square inch gage (psig) for not less than three (3) seconds applied through the detonator well cavity. Observation shall be made for leaks as evidenced by a loss of pressure to determine compliance with 3.5.

4.5.6 Pull test of body crimp of fuze assembly. The fuze assembly shall be subjected to a tensile load of fifteen (15) pounds applied between the pressure plate and the lower portion of the body. The load shall be applied at such a rate that the pull fixture moves at a speed of 1/2 to 1 inch per minute. An air pressure shall then be applied and observation made for leaks as specified in 4.5.5 to determine compliance with 3.6.

4.5.7 Recovery test of firing pin of fuze assembly. The lower portion of the body in the fuze assembly shall be supported securely and the safety fork removed. A load of one hundred thirty-four plus one pound (134 + 1 lb./ - 0 lb.) shall be applied to the fuze assembly at such a rate that the load fixture moves at a rate of 1/2 to 1 inch per minute and then released. Observation shall be made that the firing pin returns sufficiently such that the firing pin point is not less than 0.002 inch above the detonator well shoulder to determine compliance with 3.7. If a load sufficient to cause sudden reversal of the firing pin assembly is inadvertently applied and the spring snaps through, that assembly shall be replaced with another test sample. Fuzes so tested shall be used in the functioning test (see 4.5.8) or scrapped.

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4.5.8 Functioning test of fuze assembly. This test may be performed on the same assembles as those subjected to the recovery of firing pin assembly test (see 4.5.7). The lower portion of the body shall be supported securely and the safety fork removed. A load shall be applied to the pressure plate at such a rate that the load fixture moves at a rate of 1/2 to 1 inch per minute. Observation shall be made that the load at which sudden reversal of the spring occurs in one-hundred and forty pounds (140 lbs.) minimum, and two-hundred and forty pounds (240 lbs.) maximum, and the average is between one-hundred and seventy pounds (170 lbs.) and two-hundred and twenty pounds (220 lbs.) for all fuze assemblies tested to determine compliance with 3.8. Assemblies so tested shall be scrapped.

5. PACKAGING

5.1 Preservation - packaging.

5.1.1 Level B. The Mine Fuzes, Cushions (Dwg. 9349312), Body Cups (Dwg. 9349313) and Body Discs (Dwg. 9349314) shall be preserved and packaged in accordance with MIL-STD-1169.

5.2 Packing.

5.2.1 Level B. The Mine Fuzes and components referenced in 5.1.1 shall be packed for shipment in accordance with MIL-STD-1169.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

5.4 Shipping. When components from more than one lot are shipped at one time, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. The fuze assembly covered by this specification is intended for use with the M20 Practice Mine.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. (See MIL-A-48078). Equipment designs shall be submitted as required to: Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAT-I, Dover, New Jersey 07801. Request letter of submittal should state contractor, contract number, specification number, item nomenclature, and classification of defects and test paragraph number.

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6.4 Distribution of Ammunition Data Cards. Distribution of data cards shall include the following: Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAT-M, Dover, New Jersey 07801.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities are now under the cognizance of ARRADCOM.

6.6 Supersession. This specification supersedes, in part, MIL-F-11247A, dated 15 October 1963. Procurement of Fuze, Mine, AT, Practice: M604, Metal Parts, shall, henceforth, be under this specification (MIL-F-63987 (AR)).

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-F-63987(AR): Fuze, Mine, AT, Practice: M604, Metal Parts For.

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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